MANIPAL INSTITUTE OF TECHNOLOGY
BENGALURU
(A constituent unit of MAHE, Manipal)

SCHOOL OF COMPUTER ENGINEERING

NAME OF THE BTECH PROGRAM: CSE AI

Cell Detection from High-Resolution Brain Scans and GUI Design to Compare Various SoA Deep Learning-Based Image Segmentation Techniques

Synopsis

Submitted by

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1. Introduction

1.1 General Introduction to the topic:

Advancements in medical imaging have enabled the acquisition of ultra high-resolution brain scans, which play a crucial role in understanding early brain development and detecting neurological abnormalities. Accurate detection and segmentation of cells in such images is a challenging task due to high cell density, overlapping structures, and large image sizes. Deep Learning (DL) based computer vision techniques have shown significant potential in addressing these challenges.

This project focuses on developing a Graphical User Interface (GUI) that integrates multiple state-of-the-art (SoA) deep learning models for image segmentation and object detection and applying them specifically to brain cell detection from high-resolution brain scans. The system enables efficient comparison of results obtained from different models, thereby assisting researchers in selecting suitable techniques for medical image analysis.

1.2 Organization:

The project is structured into multiple interconnected modules, each handling a specific stage of the image processing pipeline. It begins with image input through a Graphical User Interface (GUI), followed by backend processing using Flask. Based on user selection, the input image is routed to one of several deep learning models for segmentation or detection. Each model processes the image according to its architecture and produces structured output such as masks, contours, or bounding boxes.

A key component of the system is its comparative analysis capability. The GUI allows users to select and execute multiple models on the same image and visualize outputs side-by-side. This organized workflow ensures clarity, reproducibility, and systematic performance evaluation. The system is designed not only for implementation but also as a reusable research framework for medical imaging experimentation.

1.3 Area of Computer Science:

This project lies at the intersection of Artificial Intelligence, Computer Vision, and Medical Image Analysis. It primarily focuses on applying deep learning-based segmentation and detection techniques to high-resolution microscopy images. Convolutional Neural Networks (CNNs), transformer-based architectures, and instance segmentation models are used to extract meaningful cellular features from complex image data.

Additionally, the project incorporates elements of Software Engineering and Human-Computer Interaction (HCI) through the development of an interactive GUI using React and backend integration using Flask. The system also involves dataset Preparation, annotation using Roboflow, and performance evaluation metrics, thereby combining AI research with practical system development.

1.4 Hardware and Software Requirements:

The hardware requirements include a modern computing system with GPU support to handle high-resolution image processing and deep learning model inference. A minimum of 8 GB RAM is recommended, although higher memory capacity improves performance when handling large microscopy images. Adequate storage is required to manage datasets and trained model weights.

On the software side, the system is developed using Python as the primary programming language. Deep learning models are implemented using frameworks such as PyTorch or TensorFlow [1]. OpenCV [5] is used for image preprocessing and post-processing operations. The frontend GUI is built using React, while Flask is used for backend API integration. The development environment can be Windows or Linux with IDE support such as VS Code or Jupyter Notebook.

2. Need for the project:

Accurate cell detection in fetal brain microscopy images is essential for neurological research and medical studies. Manual identification and segmentation of cells are time consuming, error-prone, and impractical for large datasets. Overlapping cells and high-density regions further complicated manual analysis, creating a need for automated and scalable solutions.

This project addresses the need for a unified deep learning-based framework that enables accurate cell detection and systematic comparison of multiple state-of-the-art models. Instead of relying on a single segmentation technique, the system allows researchers to evaluate and select the most suitable model based on dataset characteristics. This reduces experimental effort and improves reliability in medical image analysis research.

3. Objective(s):

The primary objective of this project is to develop a unified deep learning framework for brain cell detection and segmentation using ultra high-resolution brain microscopy images. The system aims to integrate multiple segmentation and detection models into a single platform and evaluate their performance under consistent experimental conditions.

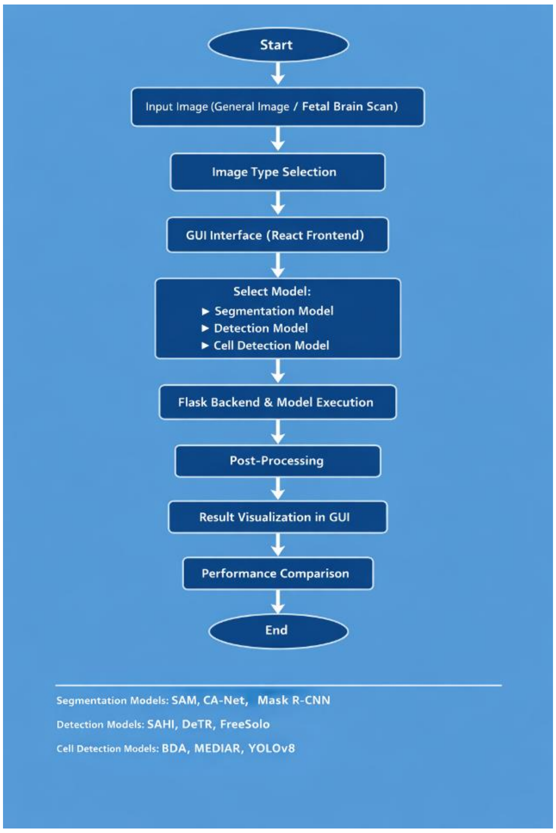
Secondary objectives include developing an interactive GUI for model selection and visualization, creating custom annotated datasets for fine-tuning YOLOv8, and enabling performance comparison using evaluation metrics such as F1-score. The project also aims to provide a reusable research tool that simplifies experimentation and accelerates model validation in medical imaging applications.

4. Methodology:

The project follows a structured methodology beginning with dataset preparation and preprocessing. High-resolution brain microscopy images are provided as input to the system. Depending on the selected model, preprocessing steps such as resizing, normalization, slicing (in SAHI), or tiling (in MEDIAR using 128×128 patches) are applied. For YOLOv8 [4] fine-tuning, images are manually annotated using Roboflow to create a structured dataset with labelled cell boundaries.

In the model execution phase, the selected deep learning architecture performs inference on the processed images. Segmentation models such as BDA, CA-Net, and Mask R-CNN [1] generate pixel-level masks, while detection models like YOLOv8 and DETR produce bounding boxes with confidence scores. MEDIAR reconstructs tiled outputs into a full segmented image. Post-processing using OpenCV refines contours and removes noise. Finally, results are visualized in the GUI, allowing side-by-side comparison and performance evaluation to determine the most suitable model for specific cellular characteristics.

Working Flow Chart:



Start: Marks the initiation of the system where the user begins the image analysis process through the application interface.

Input Image: The user uploads a high-resolution microscopy image (or general test image) that will be processed by the selected deep learning model.

Image Type Selection: The system identifies the type of input image to apply appropriate preprocessing and model configuration.

GUI Interface (React Frontend): The React-based graphical interface allows users to upload images, select models, and visualize results interactively

Select Model (Segmentation / Detection / Cell Detection): The user selects a specific deep learning model category depending on whether pixel-level segmentation or object-level detection is required.

Flask Backend & Model Execution: The backend server processes the uploaded image, routes it to the chosen model, and performs inference using the trained deep learning architecture.

Post-Processing: Model outputs are refined using image processing techniques such as contour filling, noise removal, mask reconstruction, or bounding box rendering.

Result Visualization in GUI: The processed output (segmented masks, contours, or bounding boxes) is displayed in the GUI for user interpretation and analysis.

Performance Comparison: Multiple model outputs are compared using evaluation metrics (e.g., F1-score, detection accuracy) to determine the most suitable model.

End: Represents completion of the processing cycle, where final results and comparisons are presented to the user.

5. Project Schedule:

- *January 2026*
  - Conducted literature review on state-of-the-art segmentation and detection models.
  - Studied brain microscopy datasets and understood preprocessing requirements.

- *February 2026*
  - Designed and developed the GUI using React.
  - Integrated backend using Flask for model execution.

- *March 2026*
  - Implemented and modified BDA and MEDIAR for high-resolution microscopy images.
  - Performed dataset annotation and fine-tuned YOLOv8 for cell detection

- *April 2026*
  - Conducted performance evaluation and comparative analysis of models.
  - Finalised and prepared the project report for submission and evaluation.

References:

[1] W. Abdulla, "Mask R-CNN for object detection and instance segmentation on Keras and TensorFlow," 2017. [Online]. Available: https://github.com/matterport/Mask_RCNN

[2] Meta AI, "End-to-end object detection with transformers," 2022. [Online]. Available: https://ai.meta.com/blog/end-to-end-object-detection-with-transformers/

[3] F. C. Akyon, S. O. Altinuc, and A. Temizel, "Slicing aided hyper inference and fine-tuning for small object detection," in Proc. IEEE Int. Conf. Image Processing (ICIP), 2022, pp. 966–970.

[4] F. C. Akyon et al., "SAHI: A lightweight vision library for performing large scale object detection and instance segmentation," 2021.

[5] J. Alexandre and P. Beylkin, *OpenCV 3 Computer Vision with Python Cookbook*. Packt Publishing, 2018.

PROJECT DETAILS

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